

B. Tech Degree V Semester (Supplementary) Examination, July 2009

ME 503 MECHANICS OF MACHINERY (2006 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A

(Answer all questions.)

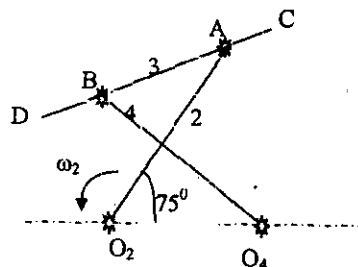
(8 x 5 = 40)

- I.
 - a. Derive the fundamental equation for correct steering.
 - b. With the help of a suitable example explain the working of an approximate straight line mechanism.
 - c. Differentiate between forward kinematics and inverse kinematics as applied to spatial mechanisms.
 - d. Explain the Overlay method used in the synthesis of mechanisms.
 - e. With the aid of a neat sketch discuss the various parts of a gear.
 - f. Write short note on Interference in gears. Also mention how interference can be avoided.
 - g. Deduce the expressions for velocity and acceleration of a follower executing simple harmonic motion.
 - h. Derive an expression to find the efficiency of a screw jack.

PART B

(15 x 4 = 60)

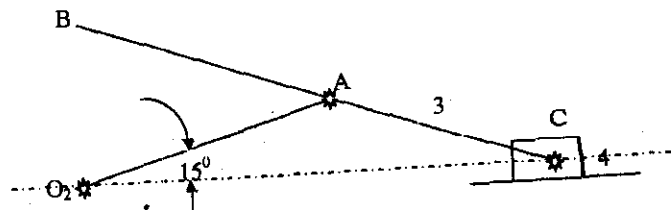
- II.
 - a. State Arnold Kennedy's three centre's in line theorem. (3)
 - b. The figure below illustrates the antiparallel or crossed -bar linkage. If the link 2 is driven at $\omega_2 = 1 \text{ rad/s}$ ccw, find the velocities of points C and D. (12)



$AO_2 = BO_4 = 300\text{mm}$
 $BA = O_4O_2 = 150\text{mm}$
 $CA = DB = 750\text{mm}$

OR

- III. For the straight line mechanism shown in the figure below,
 $\omega_2 = 20 \text{ rad/s}$ cw and $\alpha_2 = 140 \text{ rad/s}^2$ cw. Determine the velocity and acceleration of point B and the angular acceleration of link 3. (15)

 $AO_2 = CA = BA = 100\text{mm}$ 

(Turn over)



- IV. Explain the procedure adopted to find the Denavit-Hartenberg parameters for a spatial mechanism. (15)

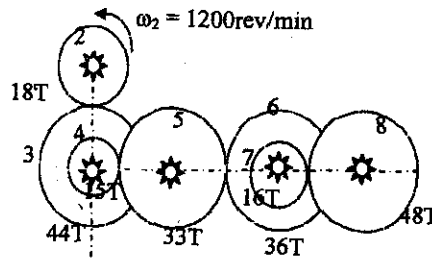
OR

- V. Adopting Freudenstein's method, synthesize a function generator to solve the equation $y = 1/x$, $1 \leq x \leq 2$ using three precision points. Take swing of input lever and output lever as 90° and starting position as 30° and 240° respectively. (15)

- VI. a. State and prove law of gearing. (5)
b. A set of gears is cast with a $17\frac{1}{2}^\circ$ pressure angle and a circular pitch of 115mm. The pinion has 20 full depth teeth. If the gear has 240 teeth, what must be its addendum in order to avoid interference? (10)

OR

- VII. a. Write short note on standard and non-standard gears. (7)
b. Find the speed and direction of gear 8 in figure. What is the speed ratio of the train. (8)



- VIII. Derive equations to describe the displacement diagram of a cam which rises with parabolic motion from a dwell to another dwell such that the total lift is L and the total cam rotation angle is β . Plot the displacement diagram and its first three derivatives with respect to cam rotation. (15)

OR

- IX. In a four-stroke petrol engine, the crank angle is 5° after t.d.c. when the suction valve opens and 53° after b.d.c. when the suction valve closes. The lift is 8mm, the nose radius 3mm and the least radius of cam 18mm. The shaft rotates at 800 rpm. The cam is of circular type with circular nose and flanks while the follower is flat faced. Determine the maximum velocity and the maximum acceleration and retardation of the valve. (15)
